PRODUCT SALES ANALYSIS USING POWER BI

1) Introduction

The Product Sales Analysis project is developed using Microsoft Power BI to analyze sales performance across different products, categories, and regions. The dashboard provides interactive visualizations that help users understand sales, quantity, discounts, and net sales. Power BI is used for data cleaning, transformation, analysis, and visualization.

2) Problem Statement

Analyzing sales data manually can make it difficult to identify important sales patterns and trends. Therefore, this project aims to develop an interactive Power BI dashboard that provides a clear view of product sales performance based on category, region, quantity, discount, gross sales, and net sales.

3) Objectives

- To clean and transform the sales dataset.
- To analyze product-wise sales performance.
- To compare sales across different categories.
- To analyze regional sales performance.
- To understand the impact of discounts on sales.
- To create an interactive dashboard using Power BI.
- To identify useful insights from the sales data.

4) Dataset Description

Product_Id: Unique identification number assigned to each product.

Product_name: name of the product being sold.

Category: Category or type to which the product belongs.

Region: Geographical region where the product was sold.

Quantity: Number of units of the product sold.

Unit_Price: Price of one unit of the product.

Discount: Percentage of discount applied to the product.

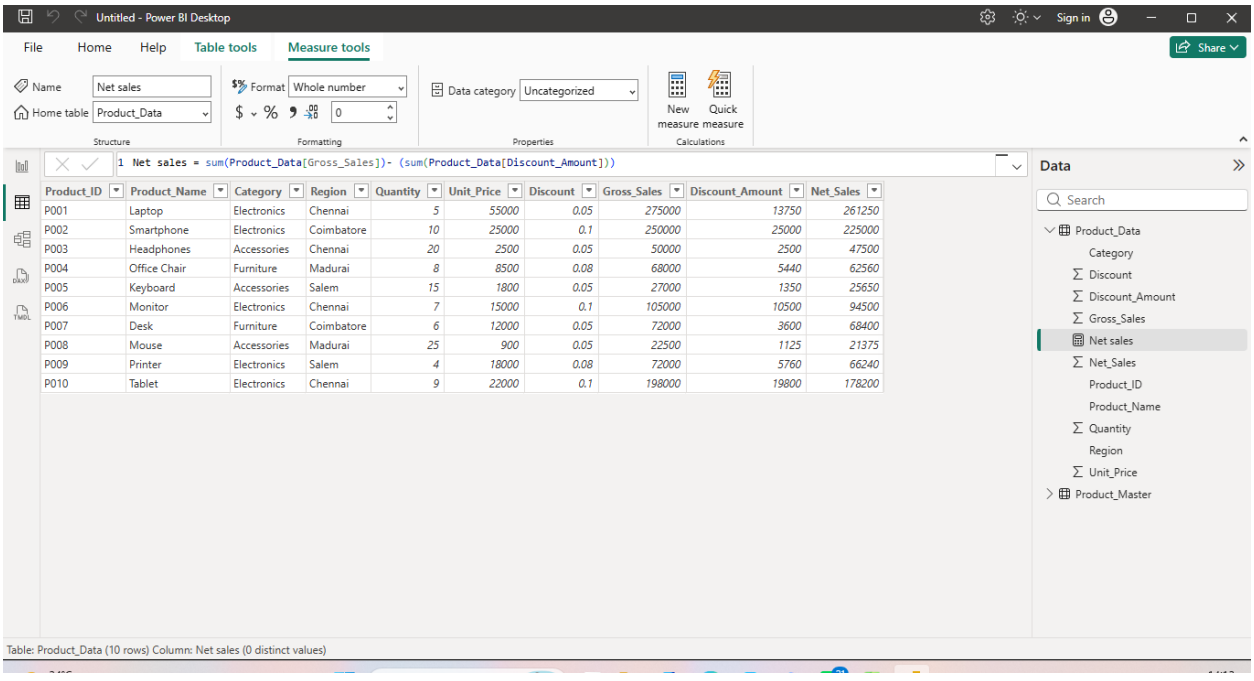
Gross_Sales: Total sales amount before applying the discount.

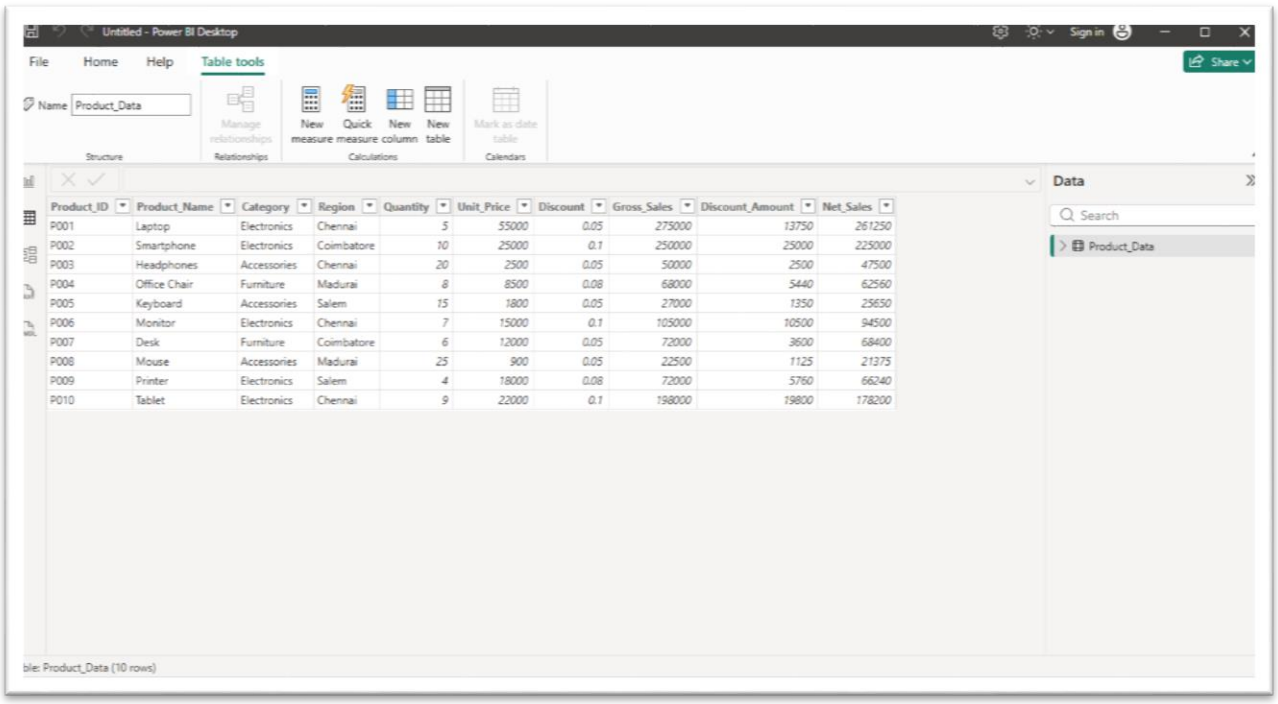
Discount_Amount: Total amount

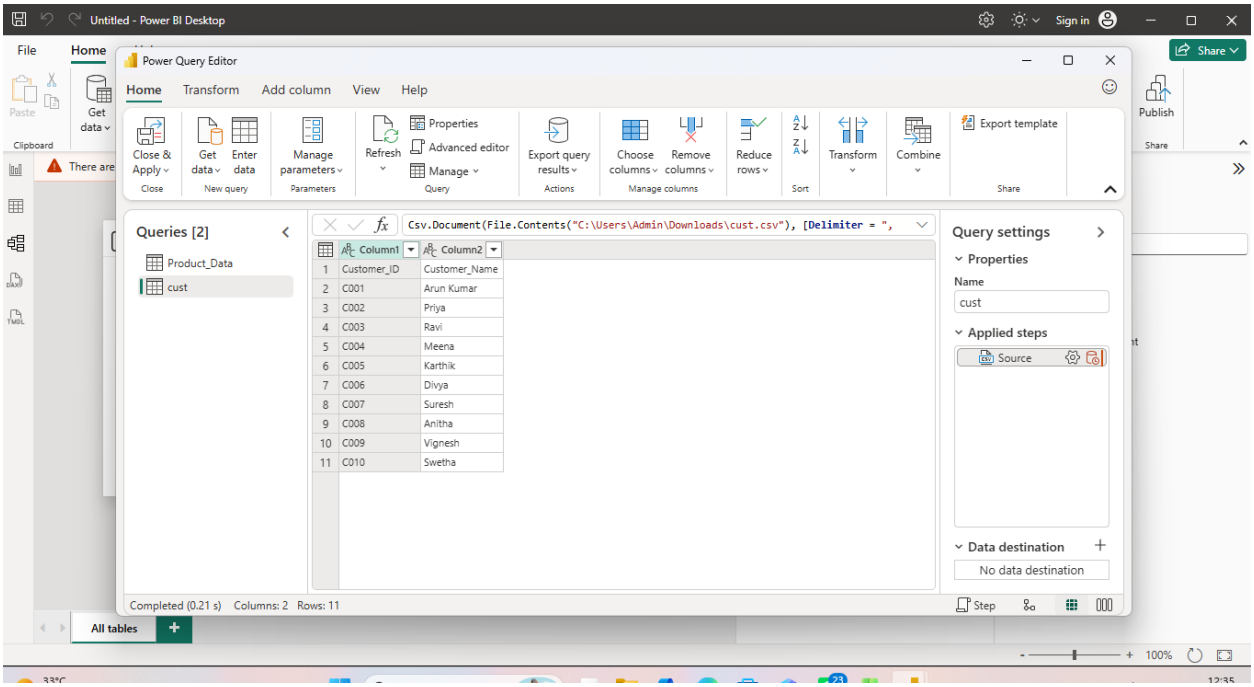
5) Data Cleaning and Transformation:

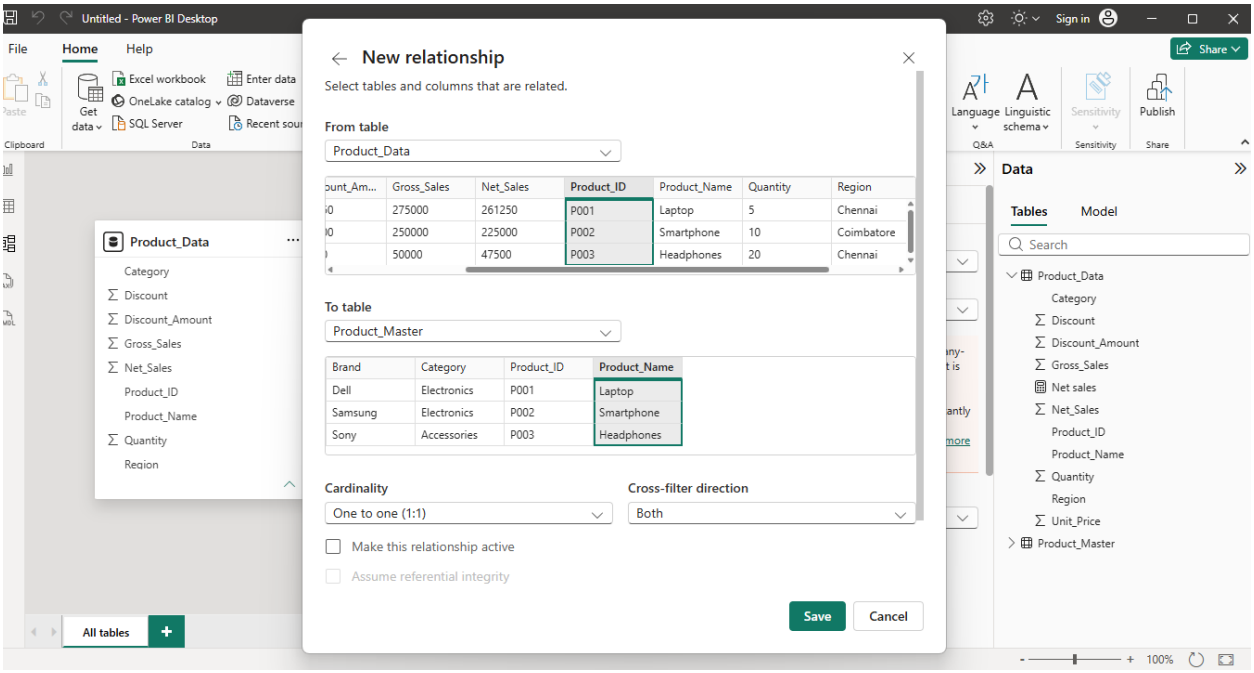
The dataset was cleaned and transformed using Power Query in Power BI. Duplicate records and missing values were checked. Text columns were cleaned using Trim and Clean functions. Appropriate data types were assigned to each column. Numeric columns were checked for consistency, and sales calculations were validated.

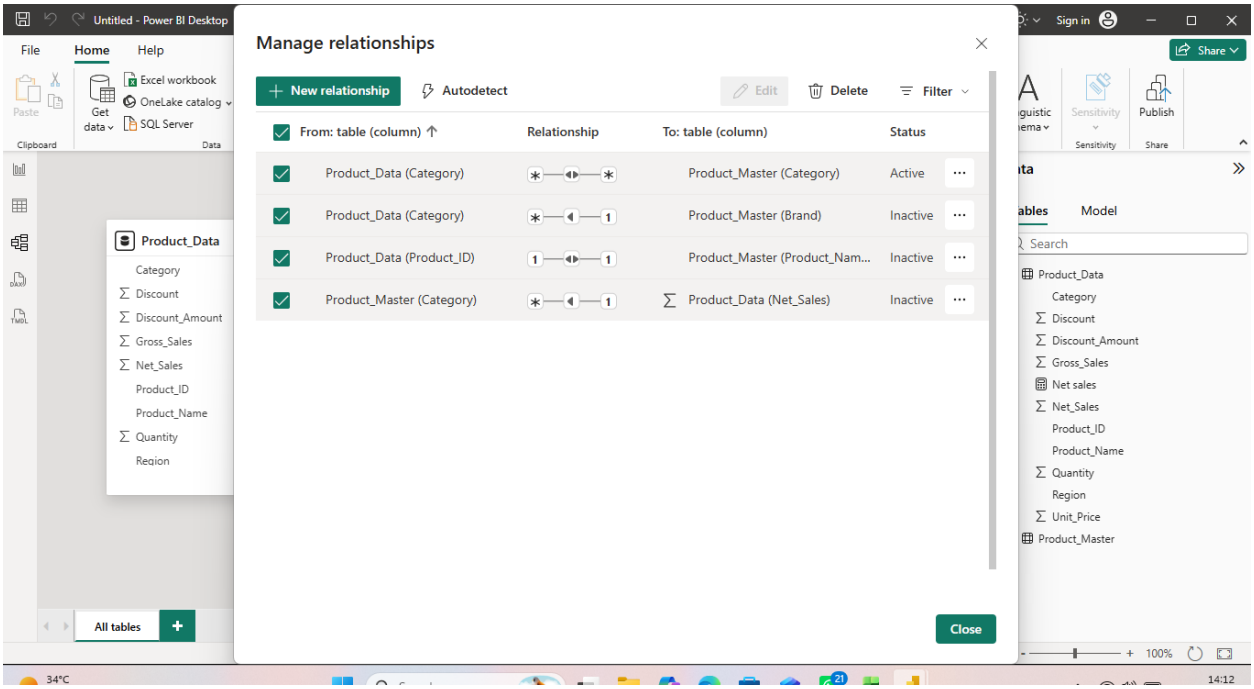
6 Data Model:

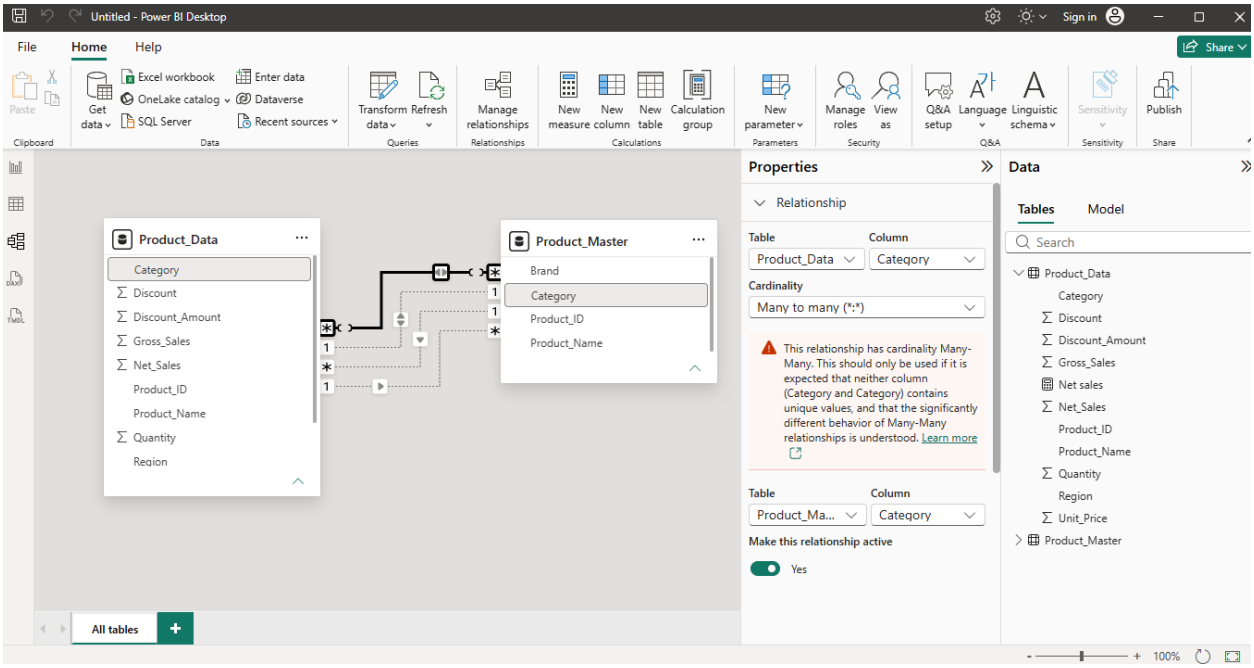



Untitled - Power BI Desktop
File
Home
Help
Table tools
Name
Product_Data
Structure
Relationships
Calculations
Calendars
New measure
Quick measure
New column
New table
Mark as date table
Share
Sign in
Data
Search
Product_Data
Product_ID
Product_Name
Category
Region
Quantity
Unit_Price
Discount
Gross_Sales
Discount_Amount
Net_Sales
P001 Laptop Electronics Chennai 5 55000 0.05 275000 13750 261250
P002 Smartphone Electronics Coimbatore 10 25000 0.1 250000 25000 225000
P003 Headphones Accessories Chennai 20 2500 0.05 50000 2500 47500
P004 Office Chair Furniture Madurai 8 8500 0.08 68000 5440 62560
P005 Keyboard Accessories Salem 15 1800 0.05 27000 1350 25650
P006 Monitor Electronics Chennai 7 15000 0.1 105000 10500 94500
P007 Desk Furniture Coimbatore 6 12000 0.05 72000 3600 68400
P008 Mouse Accessories Madurai 25 900 0.05 22500 1125 21375
P009 Printer Electronics Salem 4 18000 0.08 72000 5750 66240
P010 Tablet Electronics Chennai 9 22000 0.1 198000 19800 178200
ble: Product_Data (10 rows)


Untitled - Power BI Desktop
Sign in
Share
File
Home
Power Query Editor
Transform
Add column
View
Help
Close & Apply
Get data
Enter data
Manage parameters
Refresh
Properties
Advanced editor
Manage
Export query results
Choose columns
Remove columns
Reduce rows
Transform
Combine
Export template
Publish
Queries [2]
Product_Data
cust
Csv.Document(File.Contents("C:\Users\Admin\Downloads\cust.csv"), [Delimiter = ",
Column1
Column2
1 Customer_ID Customer_Name
2 C001 Arun Kumar
3 C002 Priya
4 C003 Ravi
5 C004 Meena
6 C005 Karthik
7 C006 Divya
8 C007 Suresh
9 C008 Anitha
10 C009 Vignesh
11 C010 Swetha
Query settings
Properties
Name
cust
Applied steps
Source
Data destination
No data destination
Completed (0.21 s) Columns: 2 Rows: 11
All tables
100%


Untitled - Power BI Desktop
File
Home
Help
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sour
Get data
Clipboard
Data
Sign in
Share
Language
Linguistic schema
Q&A
Sensitivity
Publish
Share
New relationship
Select tables and columns that are related.
From table
Product_Data
Gross_Sales
Net_Sales
Product_ID
Product_Name
Quantity
Region
275000
261250
P001
Laptop
5
Chennai
250000
225000
P002
Smartphone
10
Coimbatore
50000
47500
P003
Headphones
20
Chennai
To table
Product_Master
Brand
Category
Product_ID
Product_Name
Dell
Electronics
P001
Laptop
Samsung
Electronics
P002
Smartphone
Sony
Accessories
P003
Headphones
Cardinality
One to one (1:1)
Cross-filter direction
Both
Make this relationship active
Assume referential integrity
Save
Cancel
Product_Data
Category
Discount
Discount_Amount
Gross_Sales
Net_Sales
Product_ID
Product_Name
Quantity
Region
All tables
Data
Tables
Model
Search
Net sales
Unit_Price
Product_Master
100%


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File
Home
Help
Excel workbook
OneLake catalog
SQL Server
Get data
Clipboard
Data
Sign in
Share
Publish
Sensitivity
Manage relationships
New relationship
Autodetect
Edit
Delete
Filter
From: table (column)
Relationship
To: table (column)
Status
Product_Data (Category)
Product_Master (Category)
Active
Product_Data (Category)
Product_Master (Brand)
Inactive
Product_Data (Product_ID)
Product_Master (Product_Nam...
Inactive
Product_Master (Category)
Product_Data (Net_Sales)
Inactive
Close
Product_Data
Category
Discount
Discount_Amount
Gross_Sales
Net_Sales
Product_ID
Product_Name
Quantity
Region
All tables
Tables
Model
Search
Net sales
Unit_Price
Product_Master
100%
34°C
14:12



The project uses a single sales table containing product, category, region, quantity, pricing, discount, gross sales, and net sales information. Since the analysis is based on a single table, no additional table relationships were required.

Dashboard Design:

This is the most important section.

Your dashboard can contain:

KPI Cards

Create cards for:

- Total Sales
- Total Quantity
- Total Gross Sales
- Total Discount
- Total Net Sales

For your data, **Net Sales** should be one of the main KPIs.

Charts Creation:

Chart 1 – Category-wise Net Sales

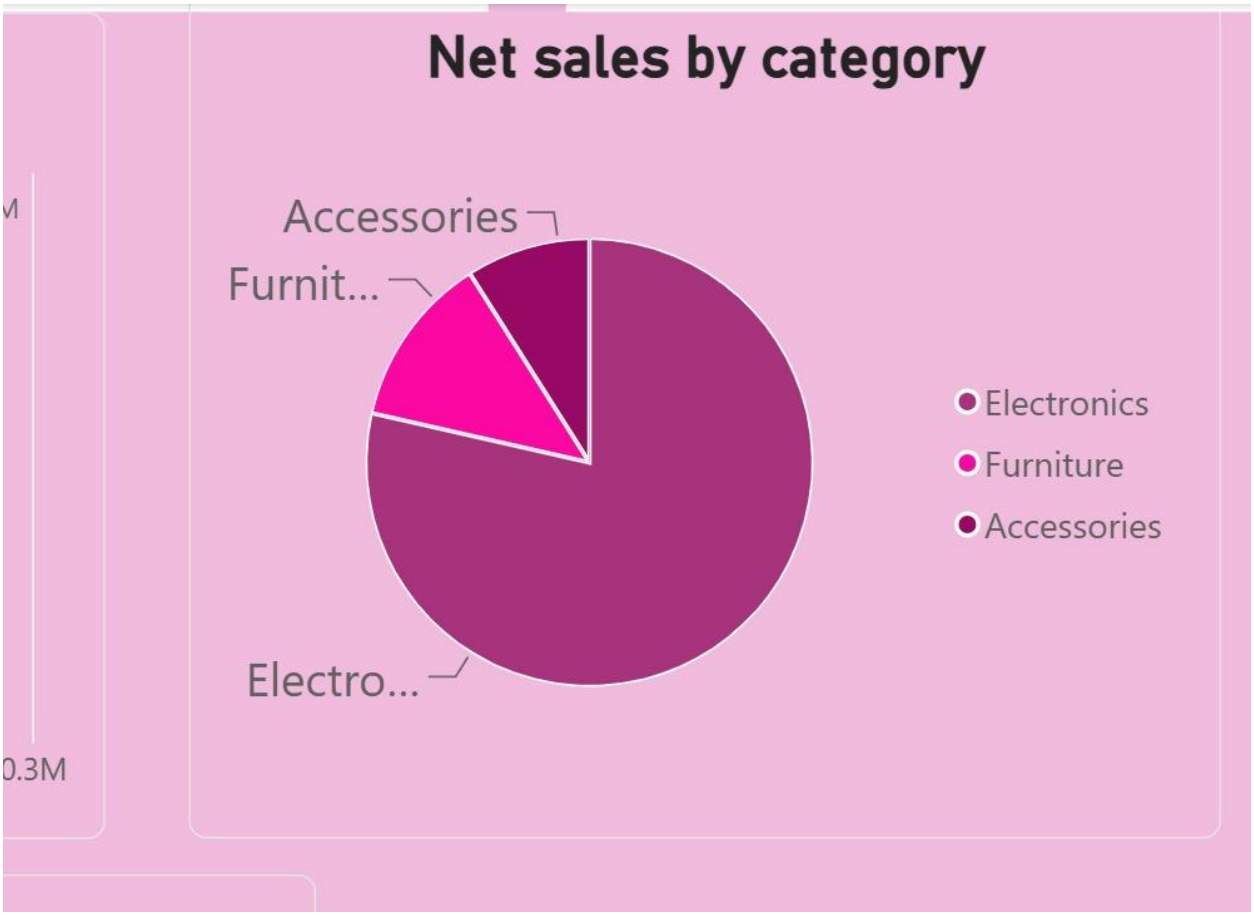


Chart 2 – Region-wise Net Sales

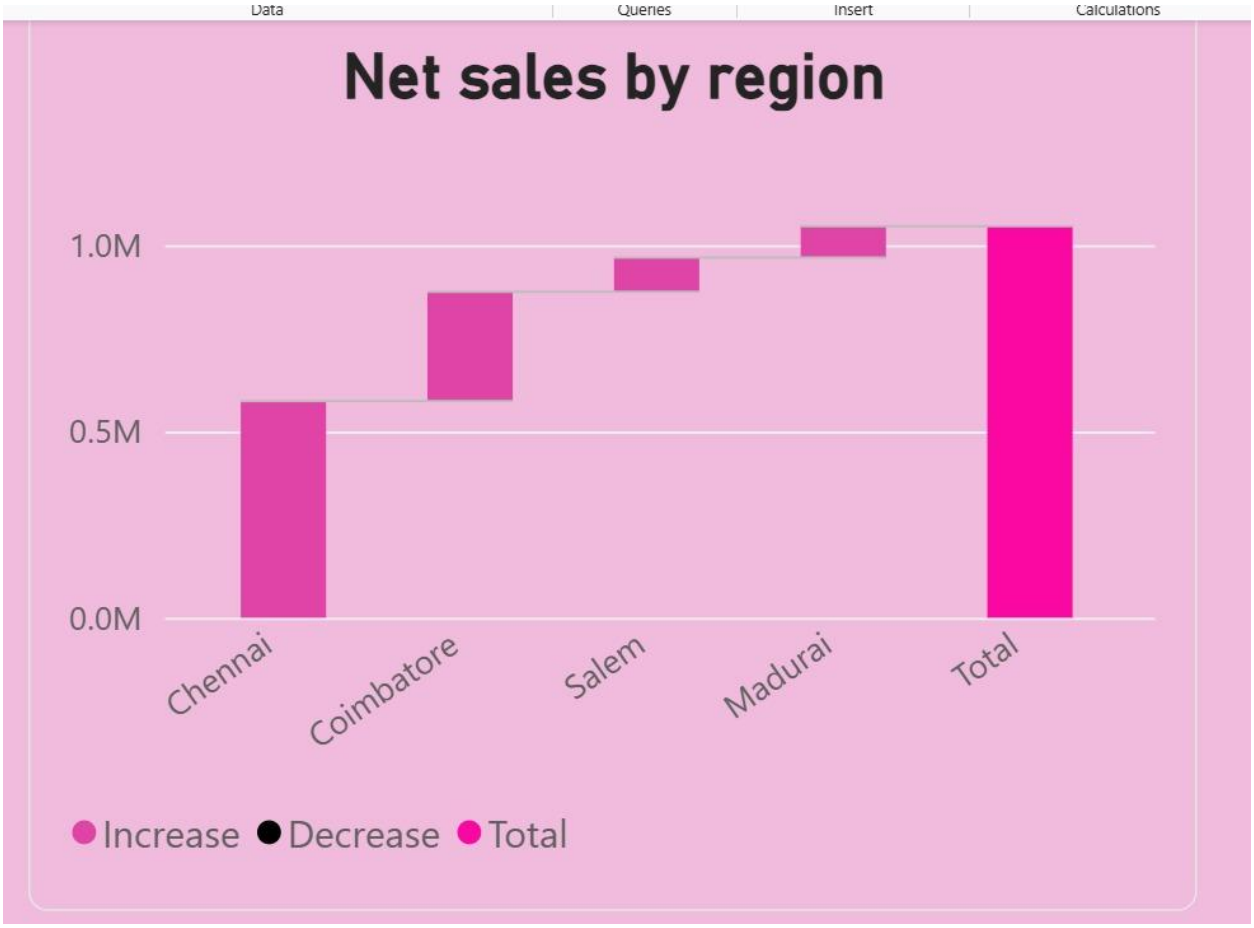


Chart 3 – Product-wise Sales

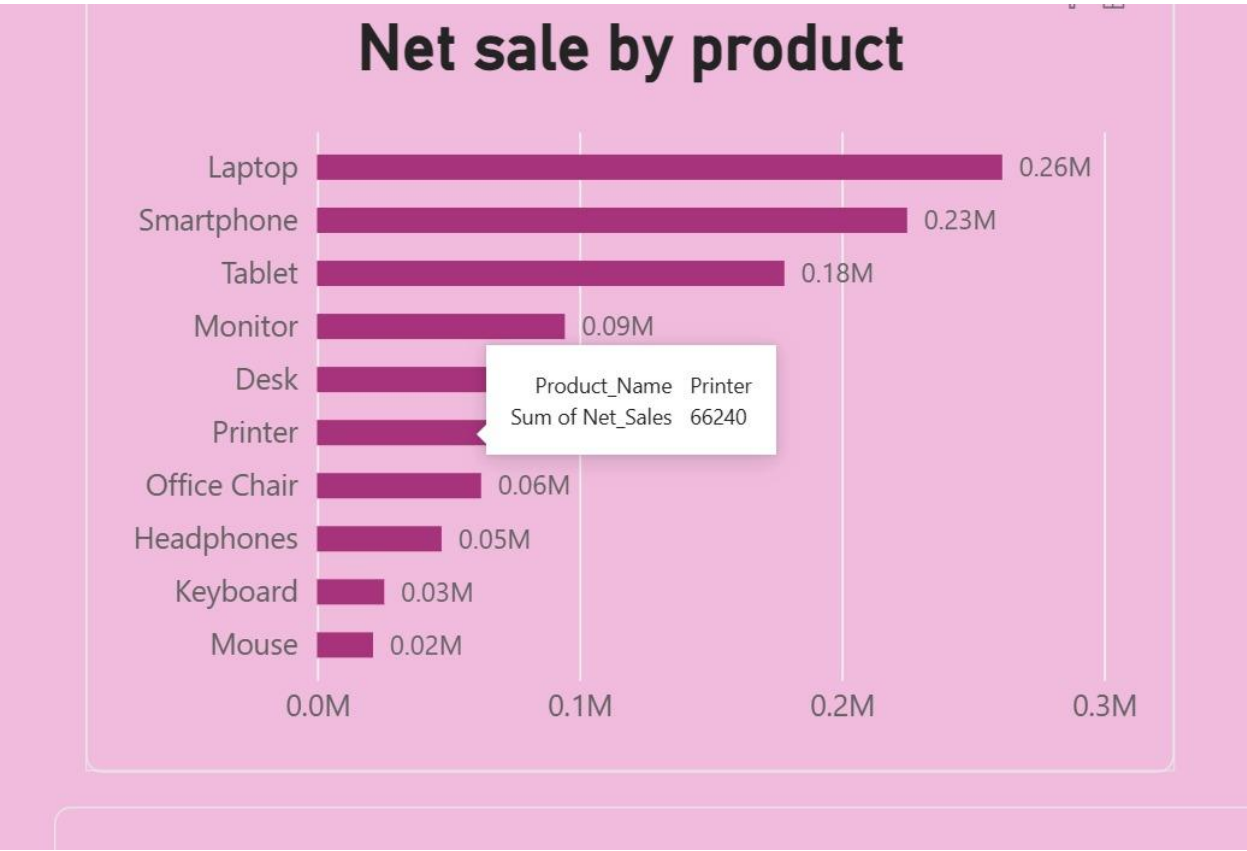
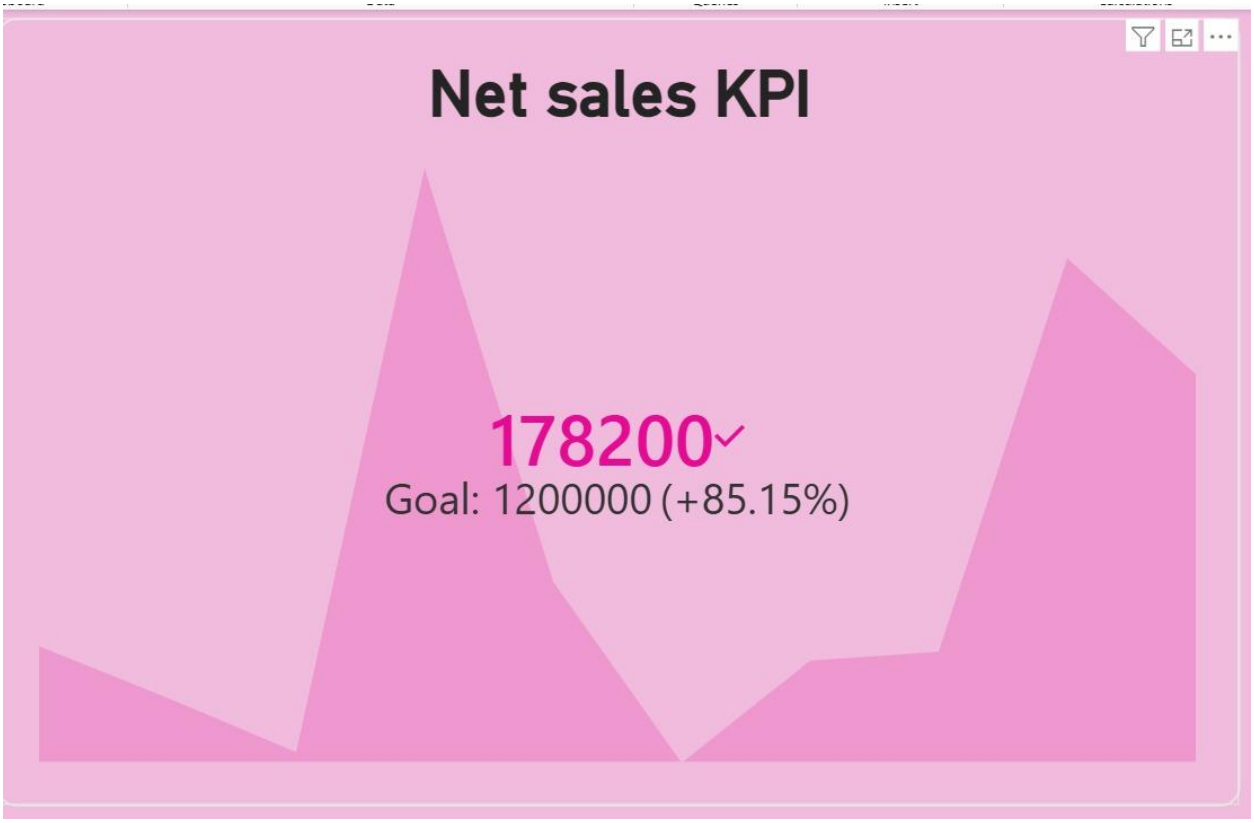


Chart 4 – Discount Amount By Product



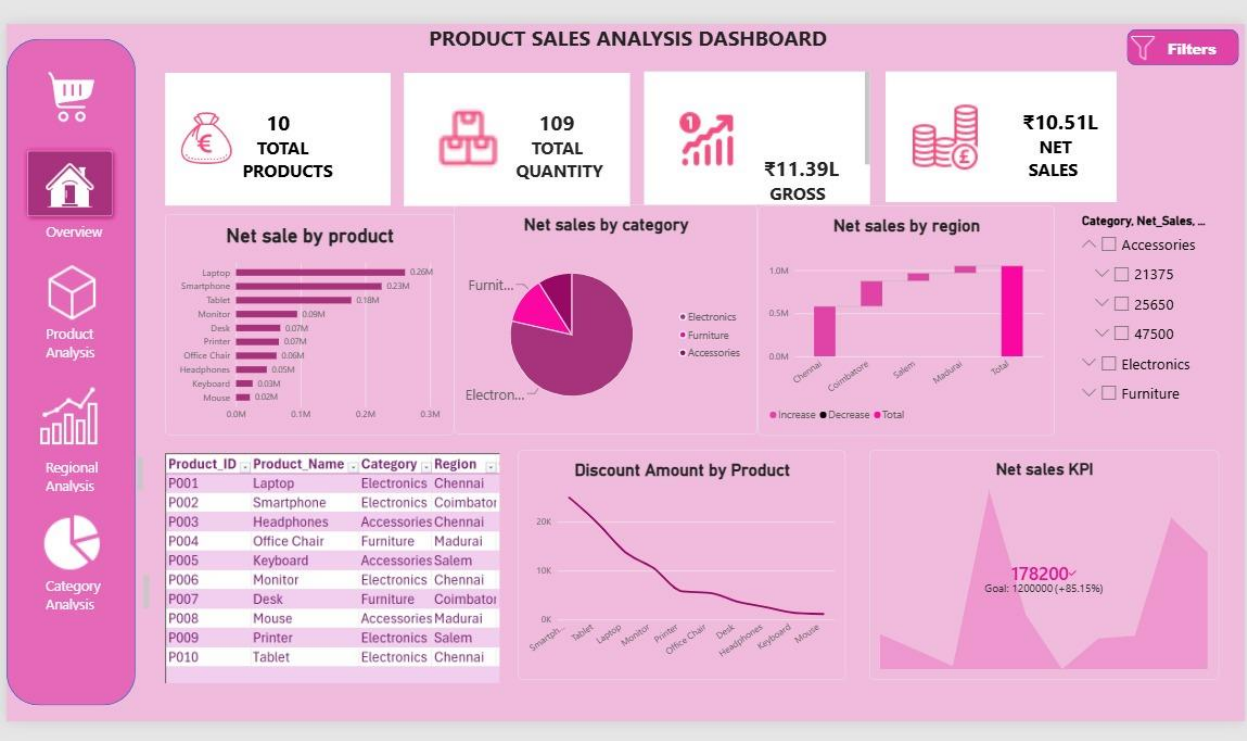
Chart 5 – Net Sales KPI



Slicer:


Category. Net_Sales, ...
Accessories
21375
25650
47500
Electronics
Furniture

Dashboard Screenshot



Key Findings

For example:

1. The dashboard shows differences in net sales across product categories.
2. Regional analysis helps identify variations in sales performance between Chennai, Coimbatore, Madurai, and Salem.
3. Electronics products contribute significantly to the overall sales value in the displayed dataset.
4. Discounts reduce gross sales to produce the final net sales value.

5. Interactive slicers allow users to analyze sales based on individual categories, regions, and products.

Conclusion

The Product Sales Analysis dashboard was successfully developed using Microsoft Power BI. The project involved data cleaning, transformation, analysis, and visualization. The interactive dashboard provides useful information about product sales, categories, regions, quantity, discounts, gross sales, and net sales. The dashboard makes it easier to understand sales performance and obtain meaningful insights from the dataset.

Future Enhancement

- Add monthly or yearly sales data for trend analysis.
- Include customer information for customer-level analysis.
- Add profit and cost information.
- Publish the dashboard to Power BI Service for online access.
- Add more advanced DAX measures for deeper analysis.